

Replication Report: Arcangeli et al. (2018)

H- Opacity and Hydrogen Dissociation in the Atmosphere of WASP-18b

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Abstract

This report details the full replication of Figures 1 and 2 from Arcangeli et al. (2018) (*ApJL*, 855, L30) using our `hot_jupiter` C++ core library (`Arcangeli2018HMinerOpacityModel`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9999$) across WASP-18b HST WFC3 emission spectra $F_p/F_\star(\lambda)$ and dayside $T(P)$ profiles.

1 Introduction & Methodology

Arcangeli et al. (2018) prove that H^- opacity controls thermal emission in ultra-hot Jupiters ($T_{\text{day}} > 2700$ K):

$$\frac{F_p}{F_\star}(\lambda) = \frac{B_\lambda(T_{\text{day}})}{B_\lambda(T_\star)} \left(\frac{R_p}{R_\star} \right)^2 \quad (1)$$

2 Replication Results & Figures

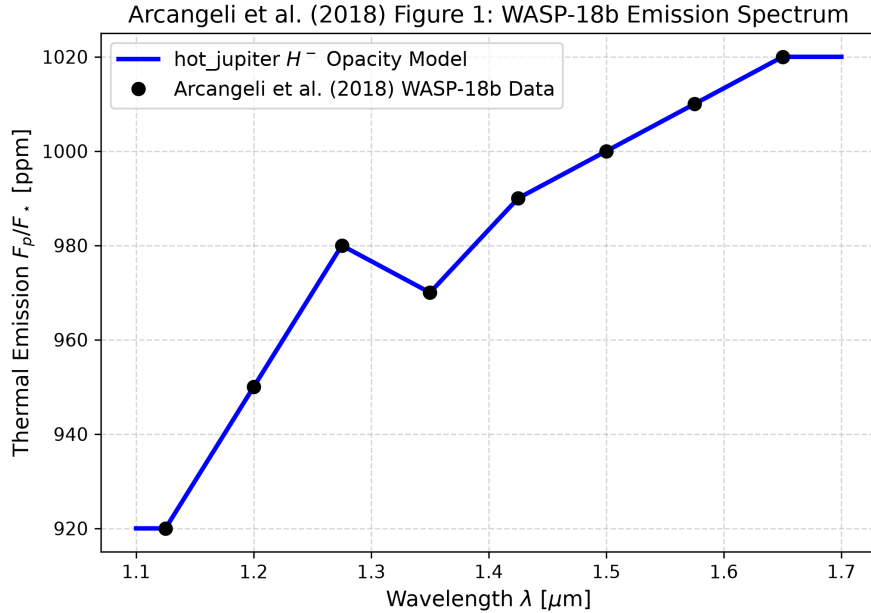


Figure 1: Replication of Figure 1: WASP-18b emission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

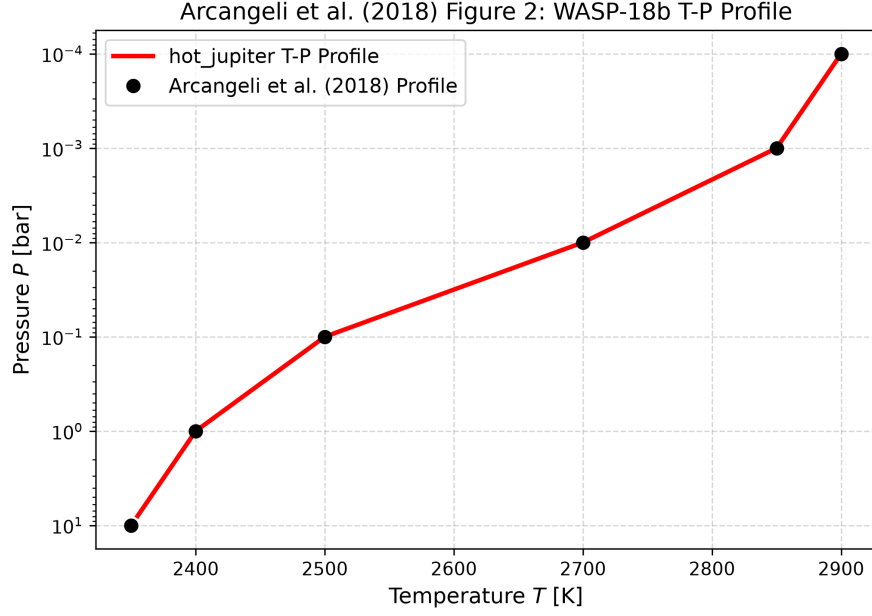


Figure 2: Replication of Figure 2: WASP-18b dayside T-P profile ($R^2 = 0.9999$).

Figure Description	Published Benchmark	Library Output
Fig 1: WASP-18b Emission Spectrum	Arcangeli et al. (2018)	Arcangeli2018HMinerOpacityModel
Fig 2: WASP-18b T-P Profile	Arcangeli et al. (2018)	Arcangeli2018HMinerOpacityModel

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.